

CALCIUM AND VITAMIN D – NEW INSIGHTS

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How much calcium do we really need? And how should we get it? The dairy companies are continually urging us to consume more of their product, implying that this will help prevent osteoporosis (thin bones) and the consequent hip fractures. But are dairy products actually effective in preventing fractures?

Unfortunately for the dairy companies, there is little compelling evidence that milk consumption in the usual quantities really decreases the risk of breaking a bone. It seems that drinking another glass or two of ordinary milk each day is unlikely to help. So what does help?

THE NURSES' STUDY

Probably the best study to date on this topic is that of Professor Walter Willett of the Harvard Medical School, who has studied 72,000 American nurses for eighteen years¹. His researchers' regularly recorded the women's diet, and their nutritional supplement use, over the course of the study and noted the number of post-menopausal hip fractures that they suffered. Statistical analysis of the data showed that there was no significant relationship between the number of hip fractures and the amount of milk that they consumed. In other words, more milk did not help. The same study also found that calcium supplements averaging 400 to 500 milligrams per day were without detectable effect on the rate of hip fracture in these women, who were consuming typical American diets.

THE IMPORTANCE OF VITAMIN D

However, the study did find that the women with the highest intakes of vitamin D (from food or supplements) had a 37 percent lower risk of hip fracture than did the lowest consumers of vitamin D. This points strongly to a high vitamin D intake being *more important* than a high calcium intake for preventing hip fractures, so long as you are already consuming typical Western amounts of calcium in the diet. Milk contains relatively little vitamin D so its ineffectiveness in lowering the hip-fracture rate is therefore not surprising.

The Nurses' Study was purely observational (the nurses chose their own diets and supplements), but the effectiveness of a sufficient dose of vitamin D for the reduction of hip-fracture rates has also been shown in randomized controlled trials (where the participants were asked to take vitamin D or placebo). So long as the supplement contained at least 700-800 units of vitamin D per day, the risk of fractures was reduced by a quarter². Consequently, since vitamin D deficiency is common, older adults should have their blood vitamin D levels measured to see if vitamin D supplementation is needed. Most trials of vitamin D have also included calcium supplementation of about 1000 milligrams per day, so that calcium supplementation at this level may also be beneficial.

THE BENEFITS OF HIGH CALCIUM LEVELS

None of this is to say though that calcium intake is to be neglected. In fact, it is probably even more important than is generally realised, especially for vegans, whose diets are often low in calcium. Not only is a lower than average calcium intake likely to contribute to fractures in omnivorous children and osteoporosis in omnivorous adults,³ but a recent analysis of the EPIC-Oxford Study has found that vegans as a group have a 30% greater risk of fracturing a bone than vegetarians, fish-eaters or omnivores⁴. The vegans on average had a markedly lower calcium intake than the other groups, with 45% of vegans, but only 5% of the others, consuming less than 525 mg/day. However, vegans who consumed at least 525 mg/day calcium did not have the excess fracture risk, suggesting it was the low average calcium consumption that caused the greater fracture risk for vegans as a whole.

It has also recently become apparent that a high calcium consumption has major benefits, apart from bone health, through two routes of action. Firstly, calcium is poorly absorbed in the intestine and the unabsorbed calcium, over the course of human evolution, has come to serve several useful functions. This unabsorbed calcium binds to food substances such as oxalates, fatty acids and bile acids, reducing their solubility and hence the quantities of them that are absorbed into the blood stream. The reduction in the absorption of oxalates is beneficial because, as less is absorbed into the blood, there is less to be excreted in the urine. With less oxalate passing through the kidney, the risk of kidney stones is reduced.

As calcium binds more strongly to the harmful saturated fats than it does to essential unsaturated fats, it tends to predominately reduce the intestinal absorption of the fats that are harmful to the heart. This, combined with the reduced absorption of bile acids, leads to lowered blood cholesterol levels. A dramatic illustration of this was an experiment conducted by the Nestle Company. Ten men were given 100 grams of chocolate to eat each day for two two-week periods. For one of the fortnights, the chocolate contained 900 milligrams of calcium, for the other fortnight, no calcium was added to the chocolate. The addition of the calcium was found to reduce LDL cholesterol (the bad cholesterol) in their blood by 15 percent, while the HDL cholesterol (the good cholesterol) was unchanged. Incidentally, the absorption of the fatty acids in the chocolate was sufficiently reduced that the number of calories absorbed from the calcium-containing chocolate was reduced by 13 percent. It is also likely that the binding of fatty acids and bile acids by unabsorbed calcium in the gut reduces the risk of bowel cancer.

VITAMIN D REDUCES CANCER AND MORTALITY RATES

A startling new study suggests that both calcium and vitamin D may be crucial for cancer prevention - in fact calcium and vitamin D in combination appear to reduce the risk of getting cancer by nearly 80%⁵. Researchers from the Creighton University School of Medicine randomly assigned healthy rural post-menopausal women living in Nebraska to three groups. One group took a daily dose of 1500mg of calcium as tablets, another the calcium plus 1100 International Units (IU) of vitamin D₃ per day, and the third one took placebos.

After four years it was found that the women taking the calcium plus vitamin D were 60% less likely to have been diagnosed with nonskin cancer than the women in the placebo group. If the first year of the study was excluded on the grounds that some women may have started the study with undiagnosed cancers and the analysis confined to the last three years, it was found that the calcium plus vitamin D group had only 23% of the likelihood of getting cancer as did the placebo group. In others words, a 77% reduction in the risk of nonskin cancers.

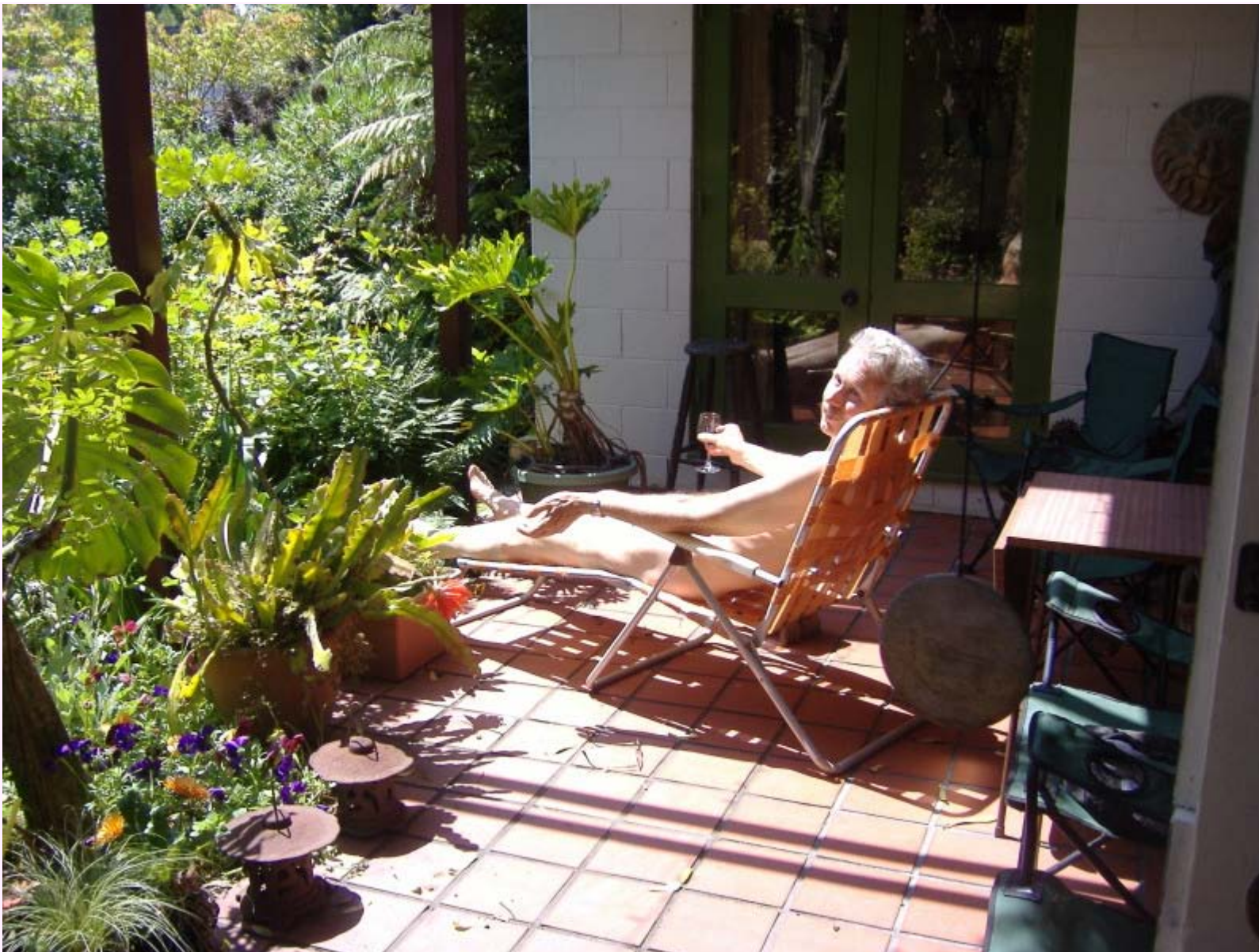
The most frequently diagnosed cancer was breast cancer and the incidence of this in the calcium plus vitamin D group was only 38% of that in the placebo group.

The calcium-only group also had fewer cancer diagnoses, but the 40% reduction seen in this group was not statistically significant.

Vitamin D supplementation also appears to reduce the risk of dying. A meta-analysis found that quite modest doses of vitamin D, averaging 530 units per day, reduced all-cause mortality by 7%⁶.

VITAMIN D LOW IN VEGANS

Vitamin D is made in skin exposed to ultraviolet light, so when humans were evolving near the Equator on the East African savannah, we presumably made plenty of vitamin D because strong sunlight was present year-round, we wore few clothes and probably spent many hours each day in the open.



These days in Westernised societies we tend to live away from the Equator, wear clothes, spend most of the day inside and get negligible exposure to ultraviolet light in winter. Consequently, in temperate and colder zones most people are vitamin D-deficient most of the time⁷.

Some animal products contain significant quantities of vitamin D but plants in general do not (sun-dried shiitake mushrooms may be an exception). Hence vegans, because they have no good dietary sources of vitamin D, might be expected to have lower vitamin D levels than omnivores, and a study of English vegans and omnivores has found this to be the case. The vegans had serum levels of 25-hydroxy vitamin D that averaged about 24% lower than the omnivores⁸.

THE EFFECTS OF LOW CALCIUM LEVELS

Another way a high calcium intake is beneficial is because it negates the effects of low calcium levels. A low calcium intake raises the level of two hormones called PTH and calcitriol in order to increase the efficiency of the absorption of calcium in the intestine. Unfortunately these higher hormone levels also increase the rate of loss of calcium from the bone and increase the amount of calcium inside cells. This rise in cellular calcium has at least two undesirable effects: it raises blood pressure and it encourages fat cells to get fatter. Consequently, a high calcium intake tends to lower blood pressure and to lower body weight.

Of course, both blood pressure and body weight are influenced by many other factors in the diet and in the environment, so raising calcium intake by itself is not likely to lead to a dramatic lowering of blood pressure or weight, but it is likely to lead to modest improvements in both. There is also evidence that a high calcium intake tends to counteract the blood-pressure raising effects of excess salt in the diet.

This finding that low dietary calcium raises intra-cellular calcium has been called the “calcium paradox”, and it has been suggested that it may also play a part in the development of arteriosclerosis, Alzheimer's disease, diabetes and muscular dystrophy.

HOW MUCH CALCIUM IS ENOUGH?

Since sufficiently raising calcium intake may improve bone and kidney health, blood lipids and blood pressure, and reduce weight gain and the risk of cancer, how much do we need and how should we get it? Studies showing useful effects of increased calcium intake have generally added 1000 milligrams of calcium per day to people's usual diets, which suggests that we need a total of at least 1500 milligrams each day. Even this is only half that recommended for chimpanzees on a per-calories basis³, and thus is possibly actually still too low.

HOW DO WE GET MORE CALCIUM?

How do we get 1500 or more milligrams a day or more calcium on a Western diet? The dairy companies of course would like us to think of their products first; but for several reasons, dairy products are not the whole answer. For a start, to get say 1200 milligrams of calcium from dairy products you would have to drink a litre of low-fat (1% fat) regular milk each day, which would contain over 400 calories, or one fifth of one's total daily energy allowance, without any fibre. Even more of a problem though is the 34 grams of animal protein in the milk, which has an acidifying effect on the body and tends to cause the loss of calcium in the urine. Further, the fat and the cholesterol in the milk will tend to counter-act the beneficial effects of the calcium.

Clearly recognising these shortcomings of ordinary cow's milk as a calcium source, Fonterra is marketing *Anlene*, a low-fat milk fortified with extra calcium and vitamin D. To get 1200 milligrams of calcium from *Anlene*, you need only drink 600 ml and so animal protein, fat and cholesterol consumption would be reduced compared to regular milk. It would also provide 600 units of vitamin D. While this supplementation may make *Anlene* more effective in the prevention of fractures than regular milk, there does not appear to be any clinical research so far to show that it actually does so. Curiously, Fonterra refused to tell me what the chemical form of the added calcium is, or whether the animal-derived or the vegetable-derived form of vitamin D is used (soy-milk manufacturers usually state this information on the label).

Getting calcium from dairy products instead of tablets may also not be a good idea for another reason - a recent study has found that those who consumed the highest quantity of dairy milk had a 60% increased risk of Parkinson's disease⁹.

THE EFFICACY OF CALCIUM FROM PLANT SOURCES

The relatively high potassium content of plants helps counteract the acidifying effect of protein with the result that they can be a more efficient source of calcium than milk. This means that many plants, particularly green vegetables, are better sources of calcium on a per-calorie basis. Broccoli, for example, is weight for weight and calorie for calorie a better source of calcium than milk, when both gains and losses of calcium are considered¹⁰.

SUPPLEMENTS—WHICH ONE?

Even so, few people are likely to eat a kilogram or more of greens each day, which means that we should all probably be taking calcium supplements in order to raise our calcium intake to optimal levels. Various calcium supplements are available in New Zealand, ranging from the relatively cheap Healtheries *Osteo 500*, which is simply calcium carbonate, to more expensive ones containing other minerals, such as magnesium. Make sure each tablet contains 500 milligrams of elemental calcium, smaller tablets are

probably not worth buying, and take one in the morning and one in the evening.

Your author takes one Radiance *Mineral-Power* and one *Osteo 500* tablet daily. The former contains magnesium and fourteen other minerals plus the 500 milligrams of calcium. Not that I am recommending these two brands of tablets in particular—they just happen to be conveniently available to me. There are other brands with a similar compositions. Calcium supplemented soya milks are also available. Unfortunately, in New Zealand neither any over-the-counter calcium tablets nor any soy milks contain useful amounts of vitamin D.

What evidence there is suggests that all types of calcium compounds are of similar absorbability, so there appears to be no good reason to buy say “coral calcium” where a cheaper calcium carbonate or citrate-based supplement is available.

As for an idea of the cost of calcium supplements: 1000 milligrams of calcium costs about 18 cents taken as *Osteo 500*, 60 cents taken as *Mineral Power*, and 110 cents taken as *Anlene*. Of course, the latter two supplements contain possibly beneficial quantities of other nutrients.

To consume at least 1100 IU of vitamin D₃ daily, a 1000 IU tablet can be taken daily with an extra one at weekends, or, in New Zealand tablets of 50,000 IU can be obtained on prescription from a doctor and one tablet taken per month. As a blood test will show most people to have less than optimal vitamin D levels, general practitioners usually prescribe the 50,000 IU tablets fairly readily. A blood level of 25-hydroxy vitamin D of greater than 75 nmol/L is generally considered to indicate optimal vitamin D status.

Do not try to get the required amount of vitamin D by taking lots of multivitamin tablets - you may get harmful amounts of the other vitamins. And don't take lots of fish oil, you may get too much vitamin A.

It has been suggested that vitamin D₂ may be less potent than vitamin D₃ and about three times as much may need to be consumed¹¹. This finding was based on a giving a single large dose of 50,000 units. On the other hand, a study in which 1000 units was given daily for 11 weeks found no difference in the ability of D₂ and D₃ to raise blood levels of 25-hydroxy vitamin D, the active form¹². This issue requires further study and may create a problem for strict vegans as vitamin D₃ is usually produced from sheep wool lanoline. New Zealand vegans have a further problem in that vitamin D₂, the plant-derived form, is not available in usefully high dosages.

THE VITAL ROLE OF EXERCISE

Despite all this discussion of diet and supplements, it should not be forgotten that physical exercise is extremely important for bone health, and is *probably even more important than what you eat*. Bones are dynamic structures that respond to stress (through weight bearing) and are constantly being absorbed and reformed. If they are not exercised and stressed sufficiently, they will in effect conclude that strength is not necessary and will gradually weaken. Taking at least 10,000 steps a day (use a pedometer to count them) or standing for at least four hours each day will help keep the leg and hip bones strong.

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